BOOTSTRAPPING AND EMPIRICAL SAMPLING DISTRIBUTIONS

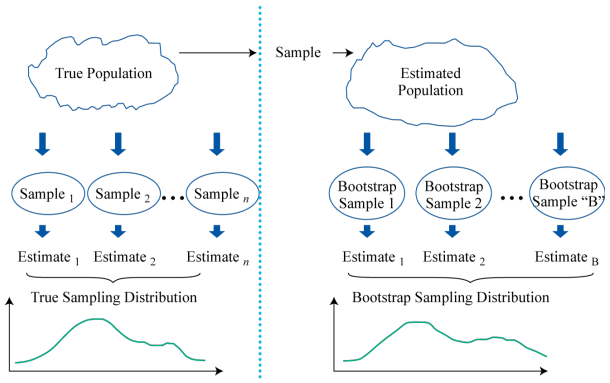
Learning Outcome

- describe the use of resampling (bootstrap, jackknife) to estimate the sampling distribution of a statistic

We demonstrated how to find the standard error of the sample mean, based on the central limit theorem. We return to the computational tool called **resampling**, which repeatedly draws samples from the original observed data sample for the statistical inference of population parameters. **Bootstrap**, one of the most popular resampling methods, uses computer simulation for statistical inference without using an analytical formula such as a *z*-statistic or *t*-statistic.

In bootstrap, we repeatedly draw samples from the original sample, and each resample is of the same size as the original sample. Note that each item drawn is replaced for the next draw (i.e., the identical element is put back into the group so that it can be drawn more than once). Assuming we are looking to find the standard error of sample mean, we take many resamples and then compute the mean of each resample. Note that although some items may appear several times in the resamples, other items may not appear at all.

Exhibit 5: Bootstrap Resampling



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Subsequently, we construct a sampling distribution with these resamples. The bootstrap sampling distribution (right-hand side of Exhibit 5) will approximate the true sampling distribution. We estimate the standard error of the sample mean using Equation 4. Note that to distinguish the foregoing resampling process from other types of resampling, it is often called model-free resampling or non-parametric resampling.

$$s_{\overline{X}} = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\hat{\theta}_b - \overline{\hat{\theta}}\right)^2}, \qquad 4$$

where:

$s_{\overline{X}}$ = the estimate of the standard error of the sample mean,

B = the number of resamples drawn from the original sample,

$\hat{\theta}_b$ = the mean of a resample, and

$\overline{\hat{\theta}}$ = the mean across all the resample means.

Bootstrap is one of the most powerful and widely used tools for statistical inference. As we have explained, it can be used to estimate the standard error of a sample mean. Similarly, bootstrap can be used to find the standard error or construct confidence intervals for the statistic of other population parameters, such as the median, which does not apply to the previously discussed methodologies. Compared with conventional statistical methods, bootstrap does not rely on an analytical formula to estimate the distribution of the estimators. It is a simple but powerful method for any complicated estimators and particularly useful when no analytical formula is available. In addition, bootstrap has potential advantages in accuracy. Given these advantages, bootstrap can be applied widely in finance, such as for historical simulations in asset allocation or in gauging an investment strategy's performance against a benchmark.

EXAMPLE 4

calculate the standard error of the sample mean. Using the bootstrap resampling method, a series of bootstrap samples, labelled as "resamples" (with replacement) are drawn from the sample of 12 returns. Notice how some of the returns from data sample in Column A feature more than once in some of the resamples (e.g., 0.055 features twice in Resample 1).

Exhibit 6: Rarely Traded Stock, 12 Monthly Returns

Column A	Resample 1	Resample 2	Resample 3		Resample 1,000
−0.096	0.055	−0.096	−0.033	…	−0.072
−0.132	−0.033	0.055	−0.132	…	0.255
−0.191	0.255	0.055	−0.157	…	0.055
−0.096	−0.033	−0.157	0.255	…	0.296
0.055	0.255	−0.096	−0.132	…	0.055
−0.053	−0.157	−0.053	−0.191	…	−0.096
−0.033	−0.053	−0.096	0.055	…	0.296
0.296	−0.191	−0.132	0.255	…	−0.132
0.055	−0.132	−0.132	0.296	…	0.055
−0.072	−0.096	0.055	−0.096	…	−0.096
0.255	0.055	−0.072	0.055	…	−0.191
−0.157	−0.157	−0.053	−0.157	…	0.055
Sample mean	−0.019	−0.060	0.001	…	0.040

Drawing 1,000 such samples, we obtain 1,000 sample means. The mean across all resample means is −0.01367. The sum of squares of the differences between each sample mean and the mean across all resample means ($\sum_{b=1}^{B}\left(\hat{\theta}_b - \overline{\hat{\theta}}\right)^2$) is 1.94143. Using Equation 4, we calculate an estimate of the standard error of the sample mean:

$$s_{\overline{X}} = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\hat{\theta}_b - \overline{\hat{\theta}}\right)^2} = \sqrt{\frac{1}{999} \times 1.94143} = 0.04408.$$

Jackknife is another resampling technique for statistical inference of population parameters. Unlike bootstrap, which repeatedly draws samples with replacement, jackknife samples are selected by taking the original observed data sample and leaving out one observation at a time from the set (and not replacing it). Jackknife method is often used to reduce the bias of an estimator, and other applications include finding the standard error and confidence interval of an estimator. According to its computation procedure, we can conclude that jackknife produces similar results for every run, whereas bootstrap usually gives different results because bootstrap resamples are randomly drawn. For a sample of size *n*, jackknife usually requires *n* repetitions, whereas with bootstrap, we are left to determine how many repetitions are appropriate.

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Question Set

1. An analyst in a real estate investment company is researching the housing market of the Greater Boston area. From a sample of collected house sale price data in the past year, she estimates the median house price of the area. To find the standard error of the estimated median, she is considering two options:

Option 1: The standard error of the sample median can be given by $\frac{s}{\sqrt{n}}$, where *s* denotes the sample standard deviation and *n* denotes the sample size.

Option 2: Apply the bootstrap method to construct the sampling distribution of the sample median, and then compute the standard error using Equation 4.

Which of the following statements is accurate?

A. Option 1 is suitable to find the standard error of the sample median.
B. Option 2 is suitable to find the standard error of the sample median.
C. Both options are suitable to find the standard error of the sample median.

Hide Solution

Solution:

B is correct. Option 1 is valid for estimating the standard error of the sample mean but not for that of the sample median, which is not based on the given formula. Thus, both A and C are incorrect. The bootstrap method is a simple way to find the standard error of an estimator even if no analytical formula is available or it is too complicated.

problems, and sampling plans frequently reflect one or more biases. The next section examines these issues.

2. Otema Chi has a spreadsheet with 108 monthly returns for shares in Marunou Corporation. He writes a software program that uses bootstrap resampling to create 200 resamples of this Marunou data by sampling with replacement. Each resample has 108 data points. Chi's program calculates the mean of each of the 200 resamples, and then it calculates that the mean of these 200 resample means is 0.0261. The program subtracts 0.0261 from each of the 200 sample means, squares each of these 200 differences, and adds the squared differences together. The result is 0.835. The program then calculates an estimate of the standard error of the sample mean.

The estimated standard error of the sample mean is closest to:

A. 0.0115
B. 0.0648
C. 0.0883

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Solution:

B is correct. The estimate of the standard error of the sample mean with bootstrap resampling is calculated as follows:

$$s_{\overline{X}} = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}(\hat{\theta}_b - \overline{\theta})^2} = \sqrt{\frac{1}{200-1}\sum_{b=1}^{200}(\hat{\theta}_b - 0.0261)^2} = \sqrt{0.004196}$$

$$s_{\overline{X}} = 0.0648$$

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OPTIONAL UNTESTED SEGMENT

UNLOCKING REAL-WORLD APPLICATIONS WITH BLOOMBERG

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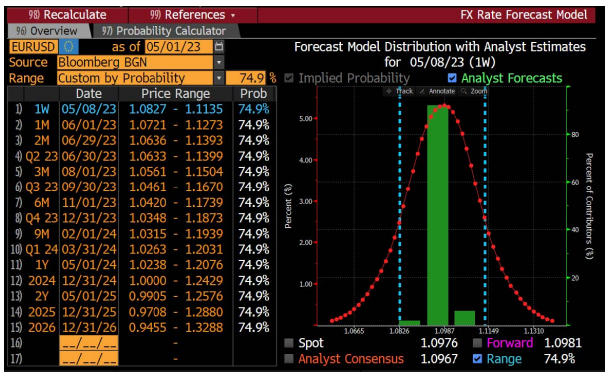
{FXFM} FX RATE FORECAST MODEL

The Bloomberg FXFM function allows you to see confidence intervals on currency pair forward values.

> "FXFM allows you to translate implied volatility levels from the FX options market into probability, so you can forecast the chances that an FX scenario will be realized on an exact date or during a selected date range. FXFM allows you to compare FX option market-based probabilities with Bloomberg analyst forecasts on a customizable chart, providing comprehensive options for generating specific FX volatility scenarios." -Bloomberg

To explore this function, type FXFM into the Bloomberg command line and press Enter or <GO>. Change the amber Range dropdown menu to "Custom by Probability" and press Enter or <GO>. Ensure that that the "Range" checkbox is selected in the bottom right of the screen. When you alter the Custom by Probability numeric amber field and press Enter, the two vertical blue bars will show the portion of the currency pair's value that lies within the selected probability portion of the center of the dataset. This will be a confidence interval as determined by your selected percentage likelihood.



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EG Regarding the first question, what does Equation 7 refer to, and in which section is it presented?

Created 6 months ago by Esteban Gómez

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PF Where exactly does 1.94143 in the bootstrap resampling example?

Created 5 months ago by Preston Fights

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